Data-driven computer animation

Tutorial 7: Position Based Dynamics

Prof. Taku Komura
TAs: Zhouyingcheng Liao(zycliao@cs.hku.hk), Mingyi Shi(myshi@cs.hku.hk)

Assignment 4

- Implement a simple cloth simulator based on position based dynamics.
- Due: 23:59, April 26th

PBD

- Position Based Dynamics (PBD) is a physics simulation framework that manipulates object positions to meet certain constraints, without direct incorporating physical principles.

- Speed: Enables real-time simulations for interactive applications such as video games and virtual reality.

- Stability: Improves stability and robustness in simulations by avoiding the direct integration of forces

The Stiffness Issue

- Real-world fabrics resist strongly to stretching, once they stretch beyond certain limits.

- But, increasing the stiffness can cause problems.
  - Explicit integrators will be *unstable*
    - Solution: smaller time steps and more computational time.
  - The linear systems involved in Implicit integrators will be *ill-conditioned*.
    - Solution: more iterations and computational time.

- Can we achieve high stiffness, with a low computational cost?

A Single Spring

- If a spring is infinitely stiff, we can treat the length as a constraint.



When the constraint is always satisfied,
the spring has the rest length
= It is infinitely stiff

$$\phi(\mathbf{x}) = \|\mathbf{x}_i - \mathbf{x}_j\| - L = 0$$

Constraint

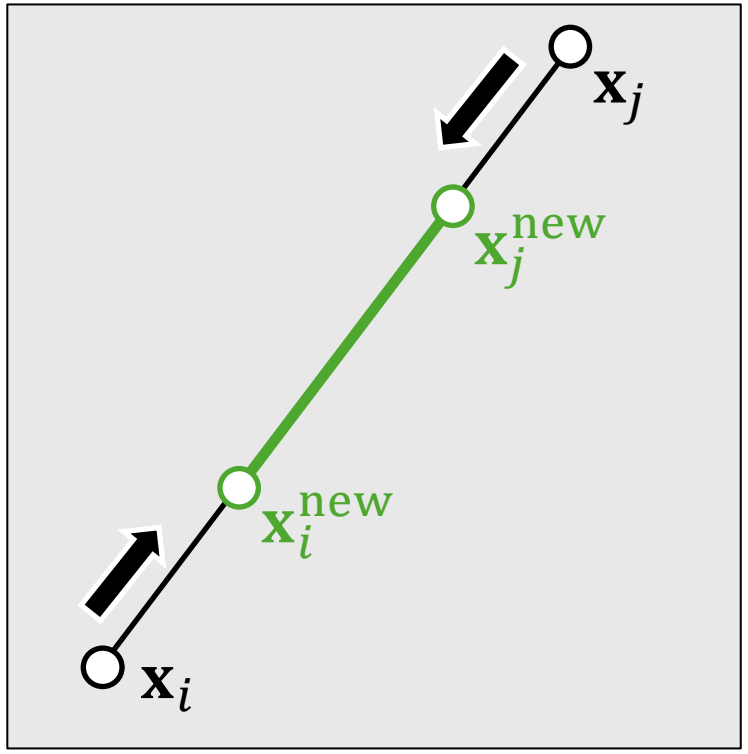
A Single Spring

- What we need now: a function that maps x to x_new so that the spring satisfies our constraint.

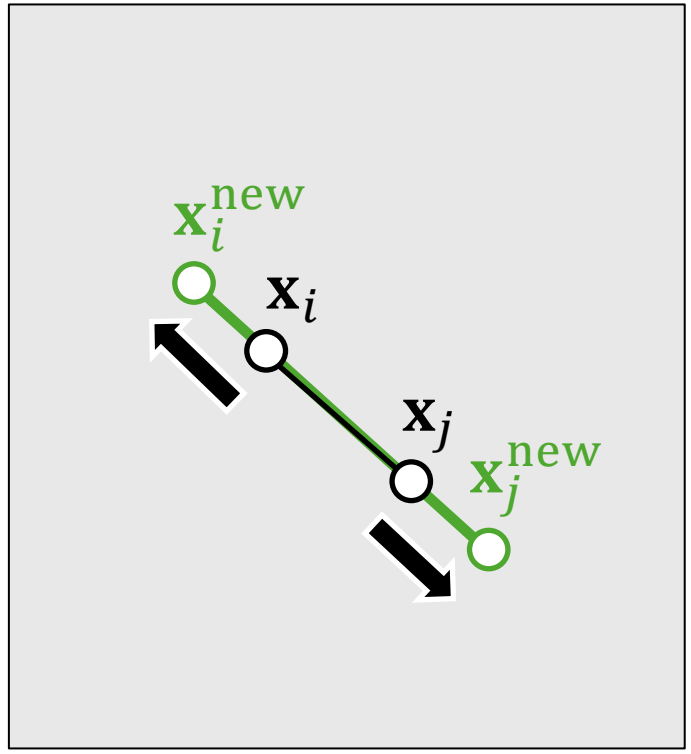


$$\phi(\mathbf{x}) = \|\mathbf{x}_i - \mathbf{x}_j\| - L = 0$$

Constraint



Stretched case



Compressed Case

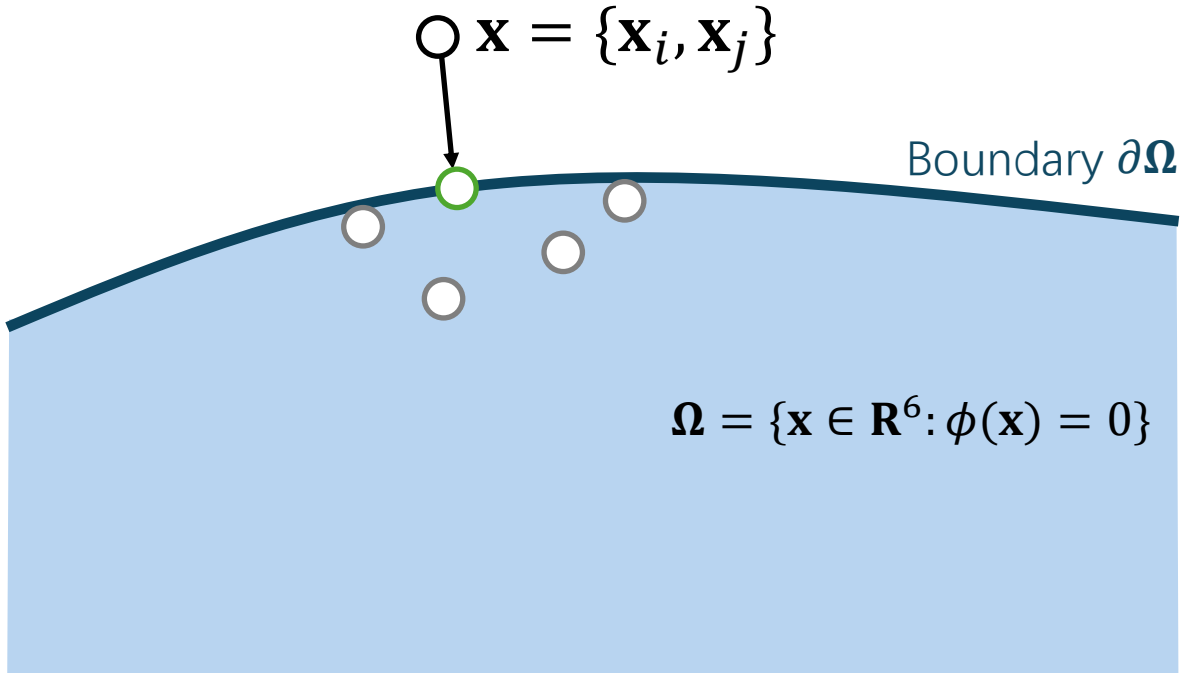
A Single Spring

- Now the problem is how to move the point into the solution space (blue region).



$$\phi(\mathbf{x}) = \|\mathbf{x}_i - \mathbf{x}_j\| - L = 0$$

Constraint



A Single Spring

- We want the moving distance to be as small as possible

  → This becomes a projection problem.

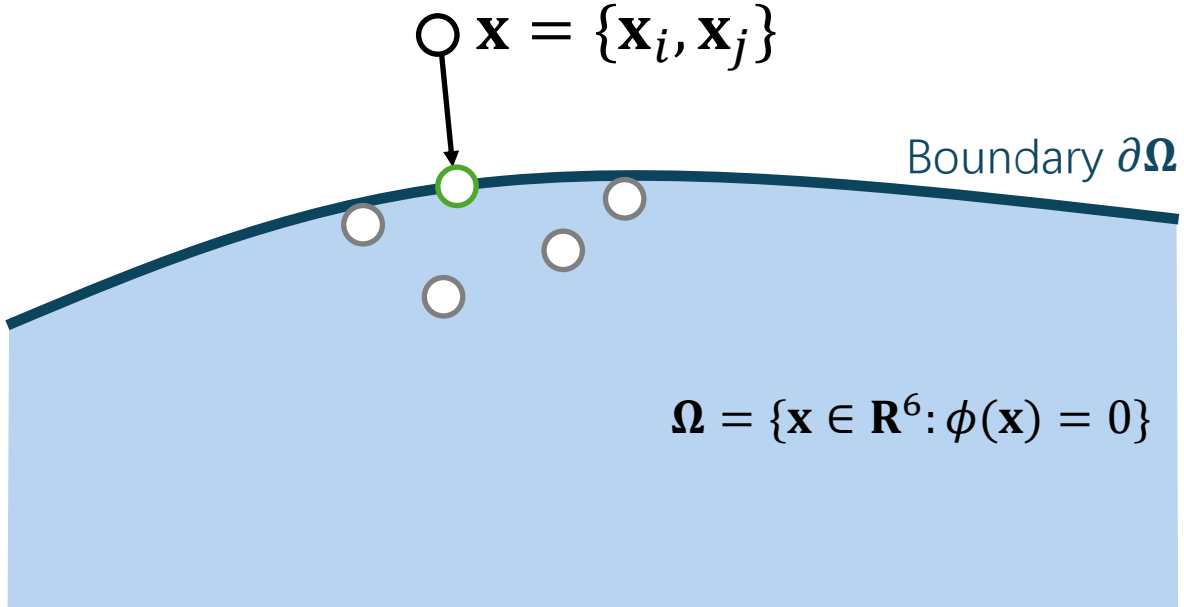
$$\{\mathbf{x}_i^{\text{new}}, \mathbf{x}_j^{\text{new}}\} = \text{argmin}\frac{1}{2}\left\{m_i\|\mathbf{x}_i^{\text{new}} - \mathbf{x}_i\|^2 + m_j\left\|\mathbf{x}_j^{\text{new}} - \mathbf{x}_j\right\|^2\right\}$$

such that $\phi(\mathbf{x}) = 0$



$$\phi(\mathbf{x}) = \|\mathbf{x}_i - \mathbf{x}_j\| - L = 0$$

Constraint



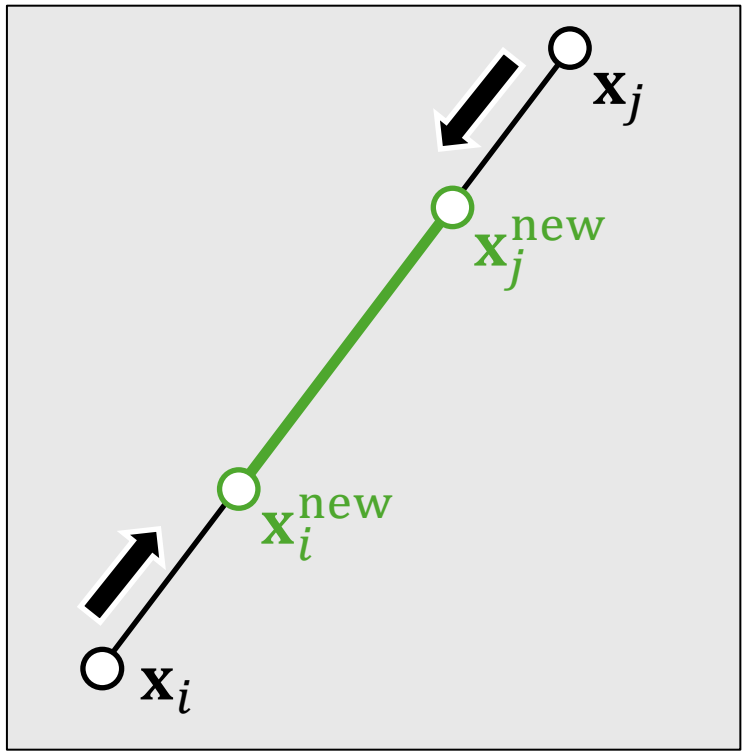
A Single Spring

- We want the moving distance to be as small as possible

  → This becomes a projection problem.

$$\{\mathbf{x}_i^{\text{new}}, \mathbf{x}_j^{\text{new}}\} = \text{argmin}\frac{1}{2}\left\{m_i\|\mathbf{x}_i^{\text{new}} - \mathbf{x}_i\|^2 + m_j\left\|\mathbf{x}_j^{\text{new}} - \mathbf{x}_j\right\|^2\right\}$$

such that $\phi(\mathbf{x}) = 0$



$$\mathbf{x}^{\text{new}} \leftarrow \text{Projection}(\mathbf{x})$$

$$\mathbf{x}_i^{\text{new}} \leftarrow \mathbf{x}_i - \frac{m_j}{m_i+m_j}\left(\|\mathbf{x}_i - \mathbf{x}_j\| - L\right)\frac{\mathbf{x}_i - \mathbf{x}_j}{\|\mathbf{x}_i - \mathbf{x}_j\|}$$

$$\mathbf{x}_j^{\text{new}} \leftarrow \mathbf{x}_j + \frac{m_i}{m_i+m_j}\left(\|\mathbf{x}_i - \mathbf{x}_j\| - L\right)\frac{\mathbf{x}_i - \mathbf{x}_j}{\|\mathbf{x}_i - \mathbf{x}_j\|}$$

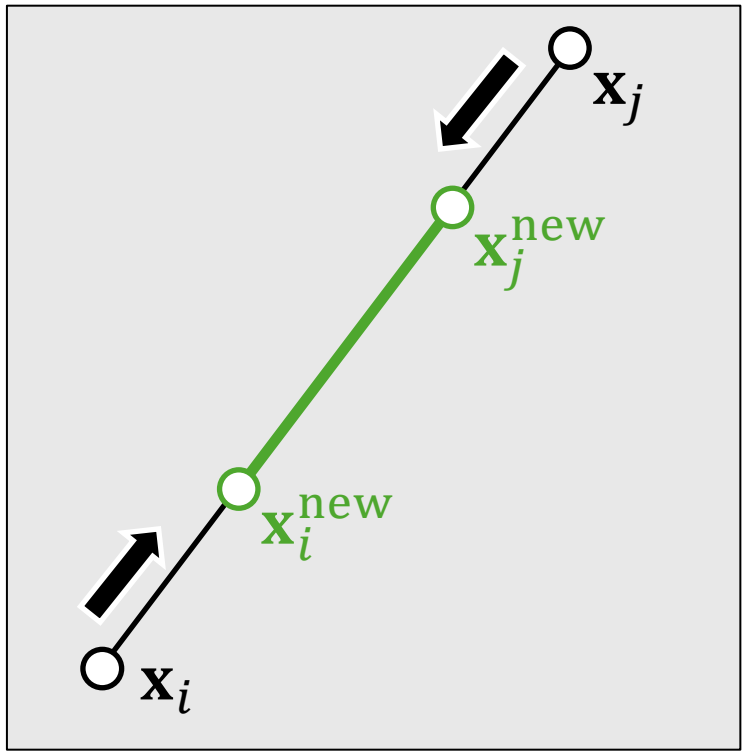
A Single Spring

- We want the moving distance to be as small as possible

  → This becomes a projection problem.

$$\{\mathbf{x}_i^{\text{new}}, \mathbf{x}_j^{\text{new}}\} = \text{argmin}\frac{1}{2}\left\{m_i\|\mathbf{x}_i^{\text{new}} - \mathbf{x}_i\|^2 + m_j\left\|\mathbf{x}_j^{\text{new}} - \mathbf{x}_j\right\|^2\right\}$$

such that $\phi(\mathbf{x}) = 0$



$$\mathbf{x}^{\text{new}} \leftarrow \text{Projection}(\mathbf{x})$$

$$\mathbf{x}_i^{\text{new}} \leftarrow \mathbf{x}_i - \frac{m_j}{m_i+m_j}\left(\|\mathbf{x}_i - \mathbf{x}_j\| - L\right)\frac{\mathbf{x}_i - \mathbf{x}_j}{\|\mathbf{x}_i - \mathbf{x}_j\|}$$

$$\mathbf{x}_j^{\text{new}} \leftarrow \mathbf{x}_j + \frac{m_i}{m_i+m_j}\left(\|\mathbf{x}_i - \mathbf{x}_j\| - L\right)\frac{\mathbf{x}_i - \mathbf{x}_j}{\|\mathbf{x}_i - \mathbf{x}_j\|}$$

The center of mass is kept

$$\phi(\mathbf{x}^{\text{new}}) = \|\mathbf{x}_i^{\text{new}} - \mathbf{x}_j^{\text{new}}\| - L = \|\mathbf{x}_i - \mathbf{x}_j - \mathbf{x}_i + \mathbf{x}_j + L\| - L = 0$$

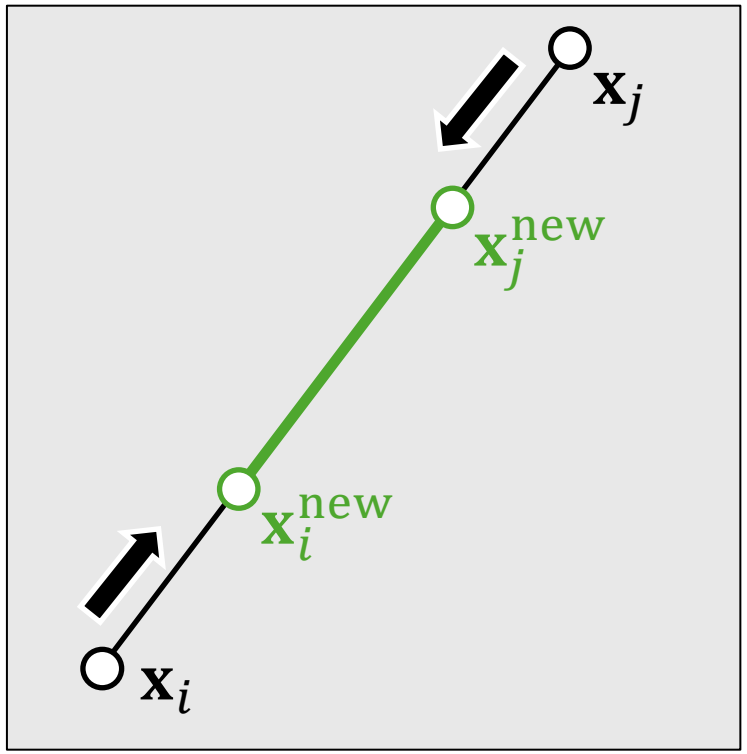
A Single Spring

- We want the moving distance to be as small as possible

  → This becomes a projection problem.

$$\{\mathbf{x}_i^{\text{new}}, \mathbf{x}_j^{\text{new}}\} = \text{argmin}\frac{1}{2}\left\{m_i\|\mathbf{x}_i^{\text{new}} - \mathbf{x}_i\|^2 + m_j\left\|\mathbf{x}_j^{\text{new}} - \mathbf{x}_j\right\|^2\right\}$$

such that $\phi(\mathbf{x}) = 0$



$$\mathbf{x}^{\text{new}} \leftarrow \text{Projection}(\mathbf{x})$$

$$\mathbf{x}_i^{\text{new}} \leftarrow \mathbf{x}_i - \frac{m_j}{m_i+m_j}\left(\|\mathbf{x}_i - \mathbf{x}_j\| - L\right)\frac{\mathbf{x}_i - \mathbf{x}_j}{\|\mathbf{x}_i - \mathbf{x}_j\|}$$

$$\mathbf{x}_j^{\text{new}} \leftarrow \mathbf{x}_j + \frac{m_i}{m_i+m_j}\left(\|\mathbf{x}_i - \mathbf{x}_j\| - L\right)\frac{\mathbf{x}_i - \mathbf{x}_j}{\|\mathbf{x}_i - \mathbf{x}_j\|}$$

When $m_i = \infty$, the particle becomes a stationary node.

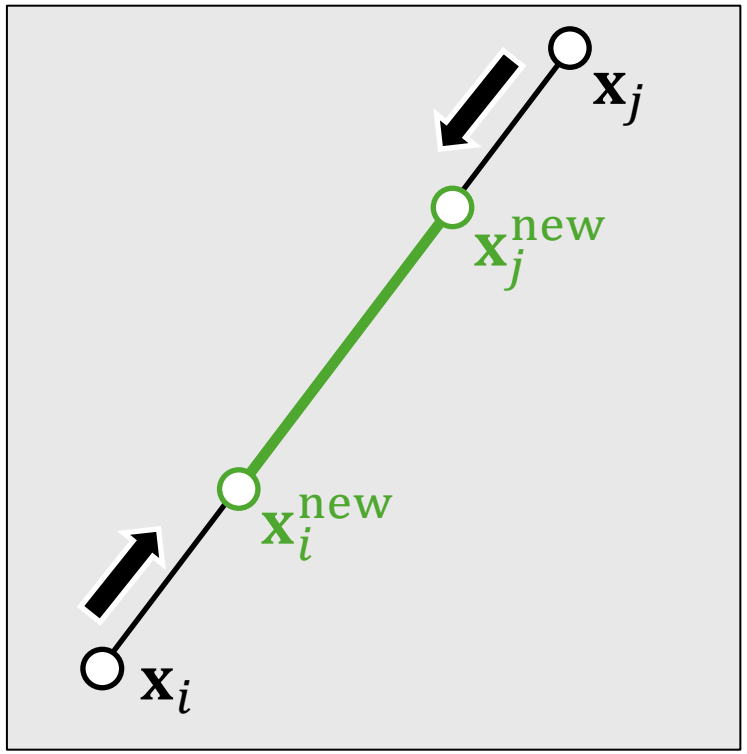
A Single Spring

- We want the moving distance to be as small as possible

  → This becomes a projection problem.

$$\{\mathbf{x}_i^{\text{new}}, \mathbf{x}_j^{\text{new}}\} = \text{argmin}\frac{1}{2}\left\{m_i\|\mathbf{x}_i^{\text{new}} - \mathbf{x}_i\|^2 + m_j\left\|\mathbf{x}_j^{\text{new}} - \mathbf{x}_j\right\|^2\right\}$$

such that $\phi(\mathbf{x}) = 0$



$$\mathbf{x}^{\text{new}} \leftarrow \text{Projection}(\mathbf{x})$$

$$\mathbf{x}_i^{\text{new}} \leftarrow \mathbf{x}_i - \frac{m_j}{m_i+m_j}\left(\|\mathbf{x}_i - \mathbf{x}_j\| - L\right)\frac{\mathbf{x}_i - \mathbf{x}_j}{\|\mathbf{x}_i - \mathbf{x}_j\|}$$

$$\mathbf{x}_j^{\text{new}} \leftarrow \mathbf{x}_j + \frac{m_i}{m_i+m_j}\left(\|\mathbf{x}_i - \mathbf{x}_j\| - L\right)\frac{\mathbf{x}_i - \mathbf{x}_j}{\|\mathbf{x}_i - \mathbf{x}_j\|}$$

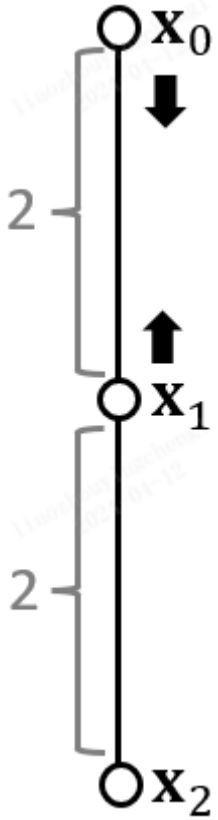
The center of mass is kept

Multiple Springs

- What about multiple springs? The Gauss-Seidel approach projects each spring sequentially in a certain order.

Multiple Springs

- What about multiple springs? The Gauss-Seidel approach projects each spring sequentially in a certain order.

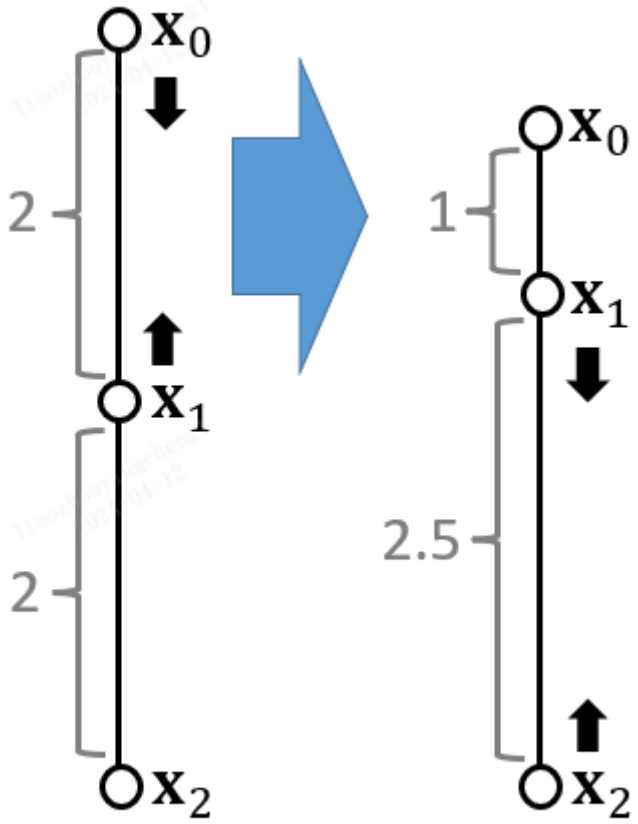


Suppose there are two springs and their rest lengths are 1.

We can start from solving the first spring

Multiple Springs

- What about multiple springs? The Gauss-Seidel approach projects each spring sequentially in a certain order.

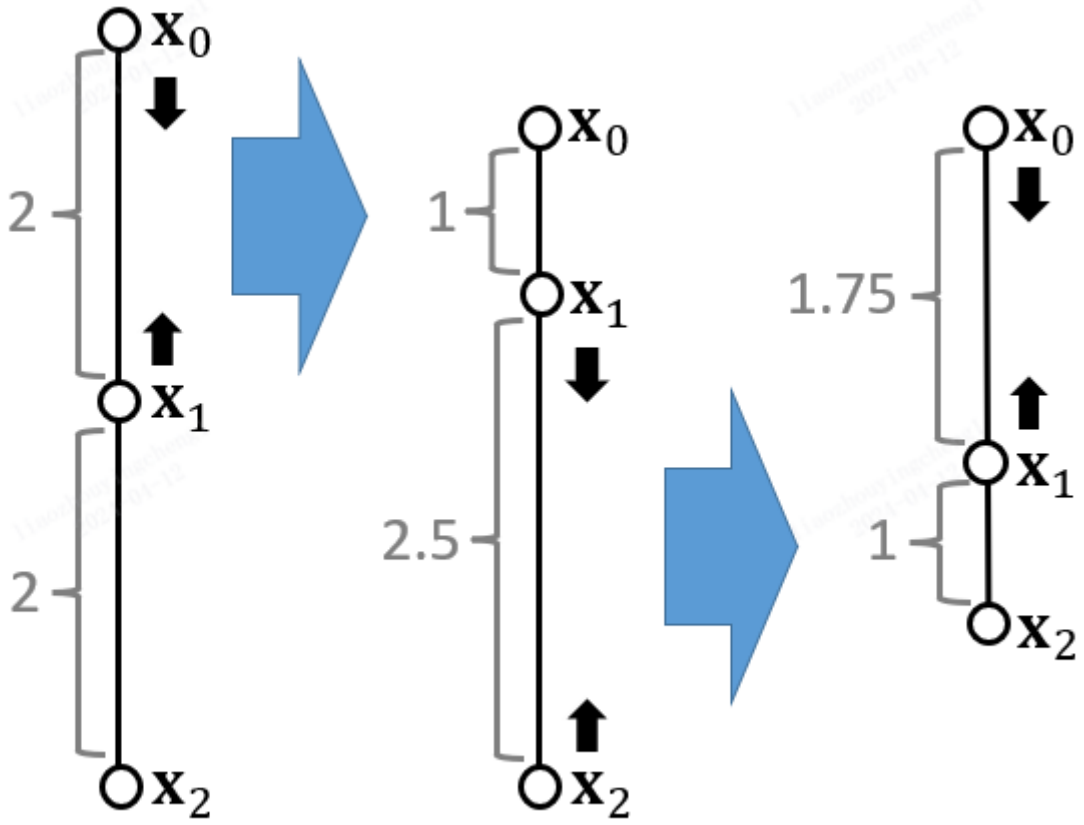


After the first spring satisfies the constraint,

We solve the second spring.

Multiple Springs

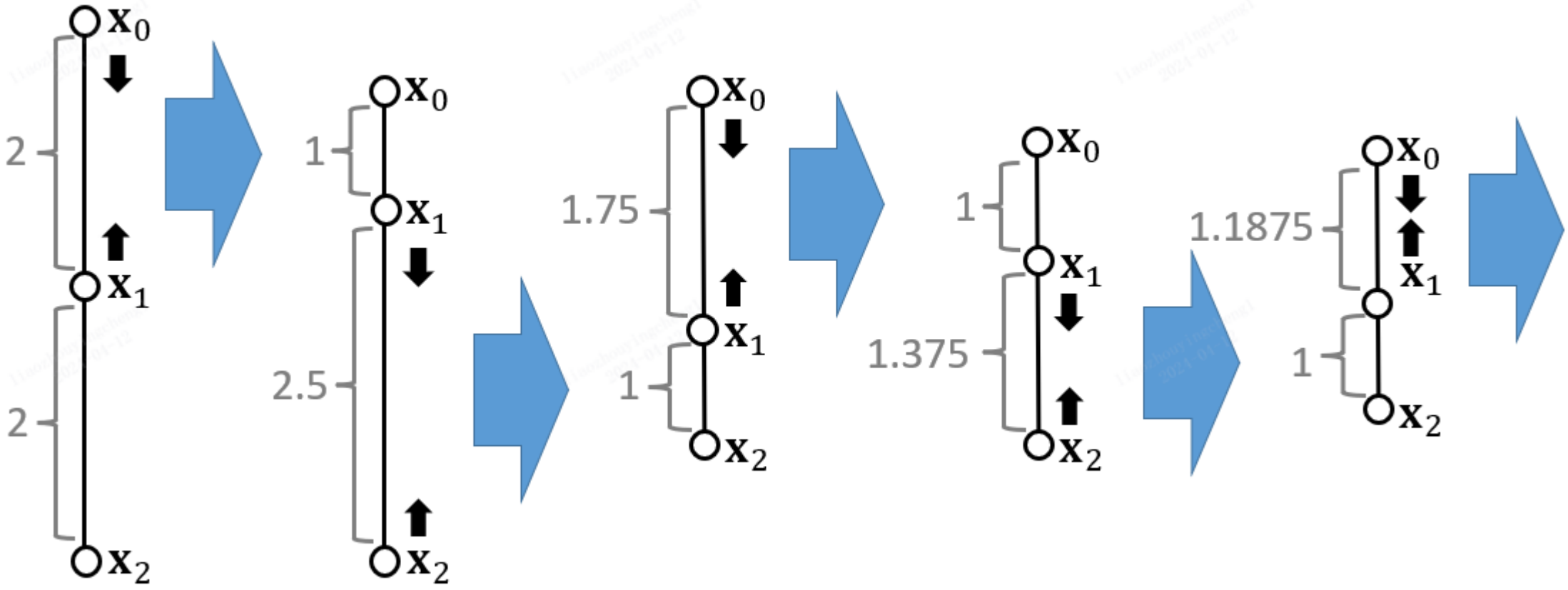
- What about multiple springs? The Gauss-Seidel approach projects each spring sequentially in a certain order.



However, when the second spring satisfies the constraint, the first spring violates it again.

Multiple Springs

- We repeat this process iteratively, and the final result will converge to the constraint.



Gauss-Seidel Approach

Projection (by Gauss-Seidel)

For $k = 0 \ldots K$

For every edge $e = \{i, j\}$

$$\mathbf{x}_i \leftarrow \mathbf{x}_i - \frac{1}{2}\left(\|\mathbf{x}_i - \mathbf{x}_j\| - L_e\right)\frac{\mathbf{x}_i - \mathbf{x}_j}{\|\mathbf{x}_i - \mathbf{x}_j\|}$$

$$\mathbf{x}_j \leftarrow \mathbf{x}_j + \frac{1}{2}\left(\|\mathbf{x}_i - \mathbf{x}_j\| - L_e\right)\frac{\mathbf{x}_i - \mathbf{x}_j}{\|\mathbf{x}_i - \mathbf{x}_j\|}$$

Gauss-Seidel Approach

Projection (by Gauss-Seidel)

For $k = 0 \ldots K$

For every edge $e = \{i, j\}$

$$\mathbf{x}_i \leftarrow \mathbf{x}_i - \frac{1}{2}\left(\|\mathbf{x}_i - \mathbf{x}_j\| - L_e\right)\frac{\mathbf{x}_i - \mathbf{x}_j}{\|\mathbf{x}_i - \mathbf{x}_j\|}$$

$$\mathbf{x}_j \leftarrow \mathbf{x}_j + \frac{1}{2}\left(\|\mathbf{x}_i - \mathbf{x}_j\| - L_e\right)\frac{\mathbf{x}_i - \mathbf{x}_j}{\|\mathbf{x}_i - \mathbf{x}_j\|}$$

- We cannot ensure the satisfaction of every constraint. But the more iterations we use, the better those constraints are satisfied.

Gauss-Seidel Approach

Projection (by Gauss-Seidel)

For $k = 0 \ldots K$

For every edge $e = \{i, j\}$

$$\mathbf{x}_i \leftarrow \mathbf{x}_i - \frac{1}{2}\left(\|\mathbf{x}_i - \mathbf{x}_j\| - L_e\right)\frac{\mathbf{x}_i - \mathbf{x}_j}{\|\mathbf{x}_i - \mathbf{x}_j\|}$$

$$\mathbf{x}_j \leftarrow \mathbf{x}_j + \frac{1}{2}\left(\|\mathbf{x}_i - \mathbf{x}_j\| - L_e\right)\frac{\mathbf{x}_i - \mathbf{x}_j}{\|\mathbf{x}_i - \mathbf{x}_j\|}$$

- We cannot ensure the satisfaction of every constraint. But the more iterations we use, the better those constraints are satisfied.
- Although the name is related to Gauss-Seidel, it differs from Gauss-Seidel.

Gauss-Seidel Approach

Projection (by Gauss-Seidel)

For $k = 0 \ldots K$

For every edge $e = \{i, j\}$

$$\mathbf{x}_i \leftarrow \mathbf{x}_i - \frac{1}{2}\left(\|\mathbf{x}_i - \mathbf{x}_j\| - L_e\right)\frac{\mathbf{x}_i - \mathbf{x}_j}{\|\mathbf{x}_i - \mathbf{x}_j\|}$$

$$\mathbf{x}_j \leftarrow \mathbf{x}_j + \frac{1}{2}\left(\|\mathbf{x}_i - \mathbf{x}_j\| - L_e\right)\frac{\mathbf{x}_i - \mathbf{x}_j}{\|\mathbf{x}_i - \mathbf{x}_j\|}$$

- We cannot ensure the satisfaction of every constraint. But the more iterations we use, the better those constraints are satisfied.
- Although the name is related to Gauss-Seidel, it differs from Gauss-Seidel.
- The order matters. The order can cause bias and affect convergence behavior.

Gauss-Seidel Approach

Projection (by Gauss-Seidel)

For $k = 0 \ldots K$

For every edge $e = \{i, j\}$

$$\mathbf{x}_i \leftarrow \mathbf{x}_i - \frac{1}{2}\left(\|\mathbf{x}_i - \mathbf{x}_j\| - L_e\right)\frac{\mathbf{x}_i - \mathbf{x}_j}{\|\mathbf{x}_i - \mathbf{x}_j\|}$$

$$\mathbf{x}_j \leftarrow \mathbf{x}_j + \frac{1}{2}\left(\|\mathbf{x}_i - \mathbf{x}_j\| - L_e\right)\frac{\mathbf{x}_i - \mathbf{x}_j}{\|\mathbf{x}_i - \mathbf{x}_j\|}$$

- We cannot ensure the satisfaction of every constraint. But the more iterations we use, the better those constraints are satisfied.
- Although the name is related to Gauss-Seidel, it differs from Gauss-Seidel.
- The order matters. The order can cause bias and affect convergence behavior.
- It's very hard to parallelize Gauss-Seidel approach. The iteration has dependency on the previous iteration.

Jacobi Approach

- Motivation
  - Reduce the bias caused by the order of edges
  - Should be easily parallelized

Jacobi Approach

- For every edge iteration, we do not modify the vertex directly.
- Instead, save the change, and update the vertex in the end using a linear blending.

Projection (by Jacobi)

For $k = 0 \dots K$

  For every vertex i

    $\mathbf{x}_i^{\mathbf{new}} \leftarrow \mathbf{0}$

    $n_i \leftarrow 0$

  For every edge $e = \{i, j\}$

    $\mathbf{x}_i^{\mathbf{new}} \leftarrow \mathbf{x}_i^{\mathbf{new}} + \mathbf{x}_i - \frac{1}{2}\left(\left\|\mathbf{x}_i - \mathbf{x}_j\right\| - L_e\right)\frac{\mathbf{x}_i - \mathbf{x}_j}{\left\|\mathbf{x}_i - \mathbf{x}_j\right\|}$

    $\mathbf{x}_j^{\mathbf{new}} \leftarrow \mathbf{x}_j^{\mathbf{new}} + \mathbf{x}_j + \frac{1}{2}\left(\left\|\mathbf{x}_i - \mathbf{x}_j\right\| - L_e\right)\frac{\mathbf{x}_i - \mathbf{x}_j}{\left\|\mathbf{x}_i - \mathbf{x}_j\right\|}$

    $n_i \leftarrow n_i + 1$

    $n_j \leftarrow n_j + 1$

  For every vertex i

    $\mathbf{x}_i \leftarrow (\mathbf{x}_i^{\mathbf{new}} + \alpha\mathbf{x}_i)/(n_i + \alpha)$

Jacobi Approach

- For every edge iteration, we do not modify the vertex directly.
- Instead, save the change, and update the vertex in the end using a linear blending.
- The problem is a lower convergence rate compared to Gauss Seidel.
- Again, the more iterations it uses, the better the constraints are enforced.

Projection (by Jacobi)

For $k = 0 \dots K$

  For every vertex i

    $\mathbf{x}_i^{\mathbf{new}} \leftarrow \mathbf{0}$

    $n_i \leftarrow 0$

  For every edge $e = \{i, j\}$

    $\mathbf{x}_i^{\mathbf{new}} \leftarrow \mathbf{x}_i^{\mathbf{new}} + \mathbf{x}_i - \frac{1}{2}\left(\|\mathbf{x}_i - \mathbf{x}_j\| - L_e\right)\frac{\mathbf{x}_i - \mathbf{x}_j}{\|\mathbf{x}_i - \mathbf{x}_j\|}$

    $\mathbf{x}_j^{\mathbf{new}} \leftarrow \mathbf{x}_j^{\mathbf{new}} + \mathbf{x}_j + \frac{1}{2}\left(\|\mathbf{x}_i - \mathbf{x}_j\| - L_e\right)\frac{\mathbf{x}_i - \mathbf{x}_j}{\|\mathbf{x}_i - \mathbf{x}_j\|}$

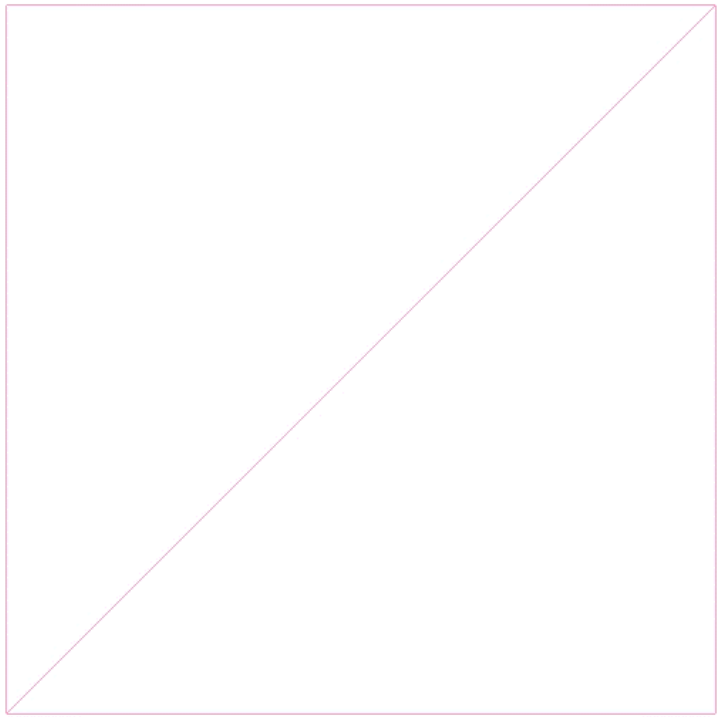
    $n_i \leftarrow n_i + 1$

    $n_j \leftarrow n_j + 1$

  For every vertex i

    $\mathbf{x}_i \leftarrow (\mathbf{x}_i^{\mathbf{new}} + \alpha\mathbf{x}_i)/(n_i + \alpha)$

Jacobi Approach



PBD Simulator

- Based on the projective approach, we can now build a PBD simulator.

A PBD Simulator

//Do Simulation, update $\mathbf{x}$ and $\mathbf{v}$

$\mathbf{v} \longleftarrow \cdots$

$\mathbf{x} \longleftarrow \cdots$

//Now PBD starts.

$\mathbf{x}^{\mathbf{new}} \longleftarrow \mathbf{Projection}(\mathbf{x})$

$\mathbf{v} \longleftarrow \mathbf{v} + (\mathbf{x}^{\mathbf{new}} - \mathbf{x})/\Delta t$

$\mathbf{x} \longleftarrow \mathbf{x}^{\mathbf{new}}$

PBD Simulator

- PBD does not have many concept of physics, e.g., stiffness coefficient, elastic force. How do we describe how much stiffness is there?

PBD Simulator

- The stiffness behavior, i.e., how tightly constraints are enforced, is subject to non-physical factors.
  - The number of iterations
  - The mesh resolution

PBD Simulator

- The stiffness behavior, i.e., how tightly constraints are enforced, is subject to non-physical factors.
  - The number of iterations
    - When there’re more iteration, the edge lengths converge to the rest lengths, i.e., it looks more stiff.
  - The mesh resolution
    - When the mesh has a lower resolution, it leads to faster convergence. Thus, with the same number of iterations, it looks more stiff.

PBD Simulator

- The stiffness behavior, i.e., how tightly constraints are enforced, is subject to non-physical factors.
  - The number of iterations
    - When there’re more iteration, the edge lengths converge to the rest lengths, i.e., it looks more stiff.
  - The mesh resolution
    - When the mesh has a lower resolution, it leads to faster convergence. Thus, with the same number of iterations, it looks more stiff.
- This method is applicable to other constraints as well, including triangle constraints, volume constraints, and collision constraints.
  - To implement these constraints, simply define their projection functions.

PBD Simulator

- Pros
  - Parallelable on GPUs
  - Easy to implement
  - Fast in low resolutions
  - Generic, can handle other coupling and constraints, including fluids
- Cons
  - Not physically correct
  - Low performance in high resolutions
    - Hierarchical approaches (can cause oscillation and other issues…)
    - Acceleration approaches, like Chebyshev

NVIDIA FleX



FleX is a particle based simulation technique for real-time visual effects.Traditionally, visual effects are made using a combination of elements created using specialized solvers for rigid bodies, fluids, clothing, etc. Because FleX uses a unified particle representation for all object types, it enables new effects where different simulated substances can interact with each other seamlessly. Such unified physics solvers are a staple of the offline computer graphics world, where tools such as Autodesk Maya's nCloth, and Softimage's Lagoa are widely used. The goal for FleX is to use the power of GPUs to bring the capabilities of these offline applications to real-time computer graphics.

Unity Plugin

Flex Github Repository

Key Features

- Artist-focused tools to ensure turnkey solutions
- Unified solver for effects
- Rigid/deformable bodies
- Phase transition
- Particles
- Fluids
- Cloth
- Rope
- Adhesion
- Gases

Platforms	Win/Linux
Dependencies	TBA
Engines	UE4 (GitHub) Unity
Links	NVIDIA FleX 1.1.0 released Siggraph 2014 paper Siggraph 2013 paper UE4 GameWorks Forum

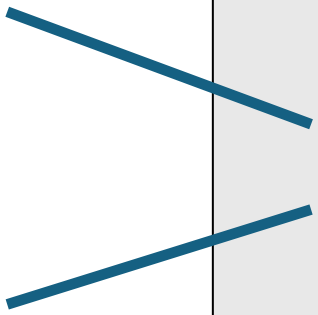
Strain Limiting

- Besides PBD, strain limiting (projective function) can be used elsewhere.

Strain Limiting

- Strain limiting aims at using the projection function for correction only.

Still do some dynamics



A Simulator with Strain Limiting

//Do Simulation, update $\mathbf{x}$ and $\mathbf{v}$

$\mathbf{v} \leftarrow \cdots$

$\mathbf{x} \leftarrow \cdots$

//Now strain limiting starts.

$\mathbf{x}^{\mathbf{new}} \leftarrow \mathbf{Projection}(\mathbf{x})$

$\mathbf{v} \leftarrow \mathbf{v} + (\mathbf{x}^{\mathbf{new}} - \mathbf{x})/\Delta t$

$\mathbf{x} \leftarrow \mathbf{x}^{\mathbf{new}}$

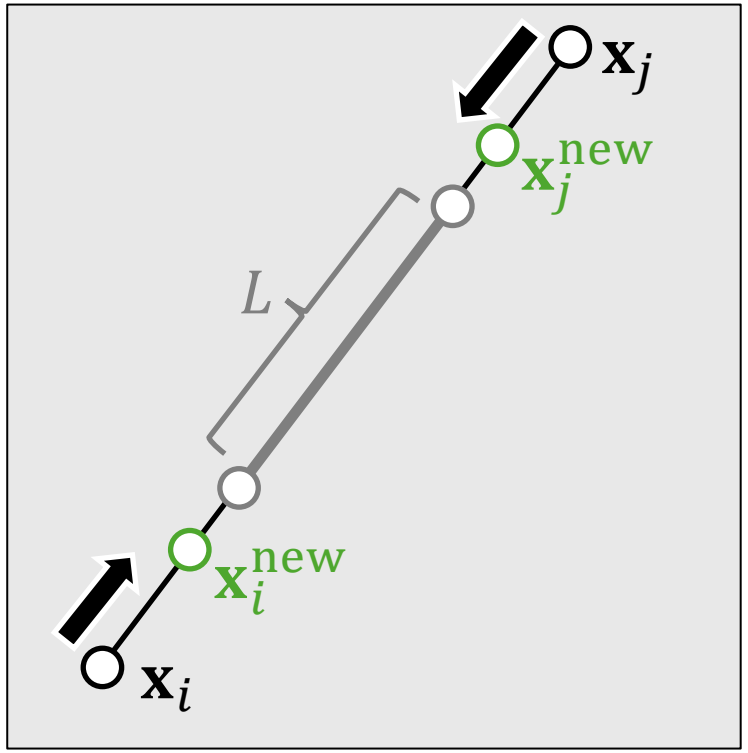
Strain Limiting

We can set the spring strain, i.e., the stretching ratio σ, to be within a limit.

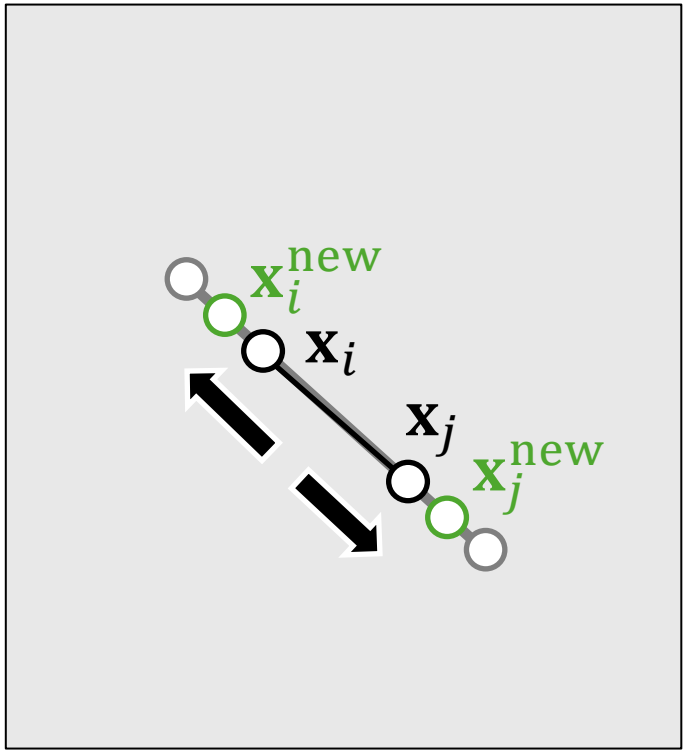


$$\sigma^{\min} \leq \frac{1}{L}\|\mathbf{x}_i - \mathbf{x}_j\| \leq \sigma^{\max}$$

Constraint



Stretched case



Compressed Case

Strain Limiting

We can set the spring strain, i.e., the stretching ratio σ, to be within a limit.



$$\sigma^{\min} \le \frac{1}{L}\|\mathbf{x}_i - \mathbf{x}_j\| \le \sigma^{\max}$$

Constraint

$$\mathbf{x}^{\text{new}} \leftarrow \text{Projection}(\mathbf{x})$$

$$\sigma \leftarrow \frac{1}{L}\|\mathbf{x}_i - \mathbf{x}_j\|$$

$$\sigma_0 \leftarrow \min\left(\max\left(\sigma, \sigma^{\min}\right), \sigma^{\max}\right)$$

$$\mathbf{x}_i^{\text{new}} \leftarrow \mathbf{x}_i - \frac{m_j}{m_i+m_j}\left(\|\mathbf{x}_i - \mathbf{x}_j\| - \sigma_0 L\right)\frac{\mathbf{x}_i - \mathbf{x}_j}{\|\mathbf{x}_i - \mathbf{x}_j\|}$$

$$\mathbf{x}_j^{\text{new}} \leftarrow \mathbf{x}_j + \frac{m_j}{m_i+m_j}\left(\|\mathbf{x}_i - \mathbf{x}_j\| - \sigma_0 L\right)\frac{\mathbf{x}_i - \mathbf{x}_j}{\|\mathbf{x}_i - \mathbf{x}_j\|}$$

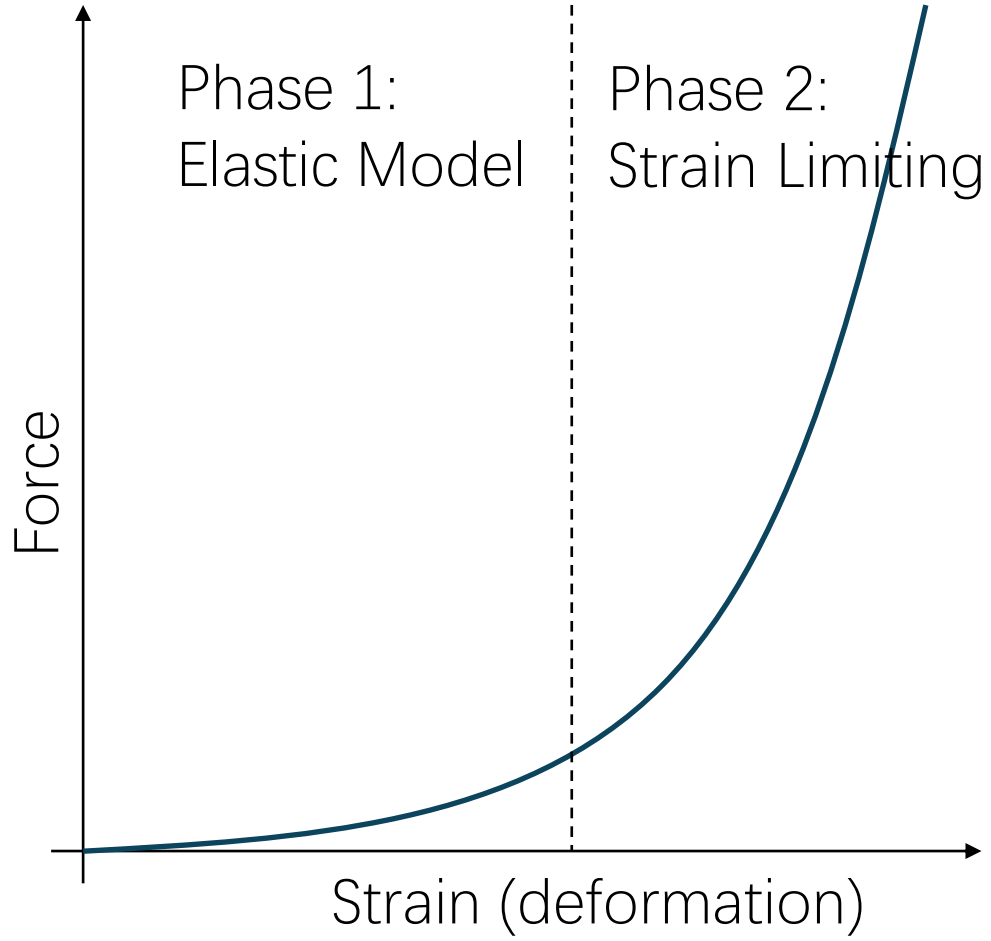
PBD: $\sigma_0 \equiv 1$;

No limit: $\sigma^{\min}, \sigma^{\max} \leftarrow \infty$

Strain Limiting in Simulation

- Strain limiting is widely used in physics-based simulation, typically for avoiding instability and artifacts due to large deformation.

- Strain limiting is useful for nonlinear effects, in a biphasic way.

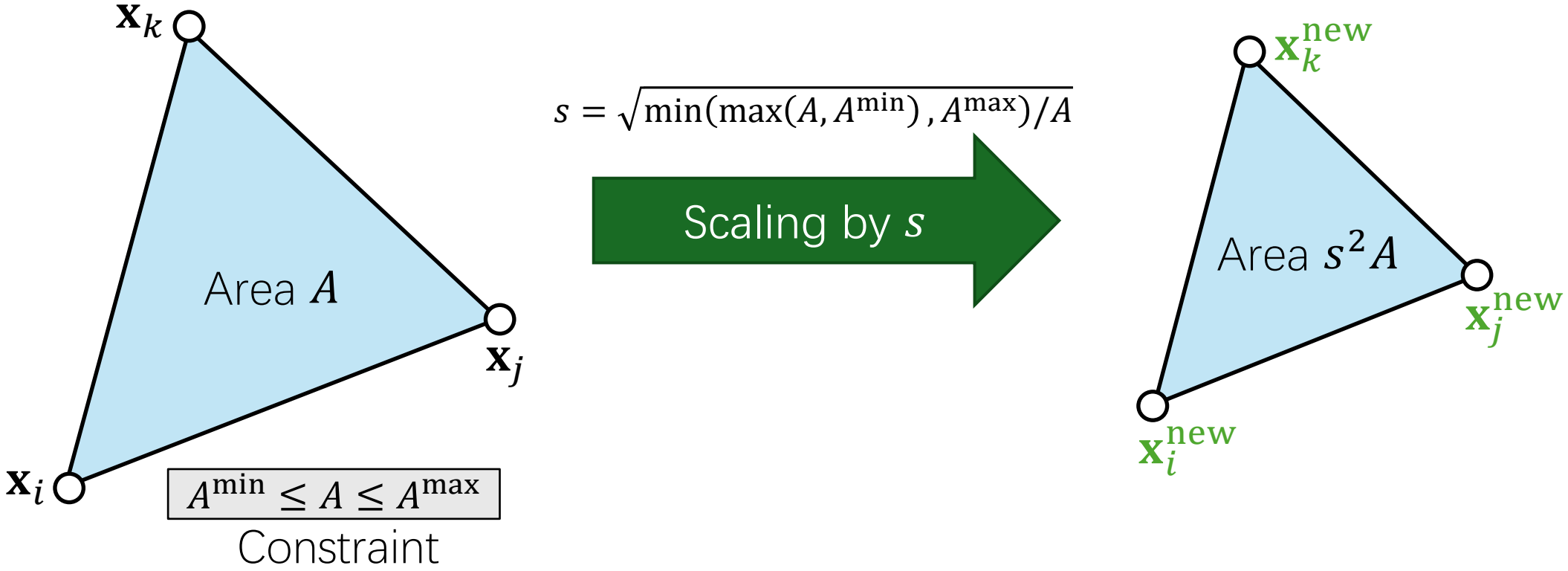


Triangle Area Limit

We can limit the triangle area as well. To do so, we define a scaling factor.

$$\{\mathbf{x}_i^{\text{new}}, \mathbf{x}_i^{\text{new}}, \mathbf{x}_k^{\text{new}}\} = \text{argmin} \frac{1}{2}\left\{ m_i \|\mathbf{x}_i^{\text{new}} - \mathbf{x}_i\|^2 + m_j \left\|\mathbf{x}_j^{\text{new}} - \mathbf{x}_j\right\|^2 + m_j \|\mathbf{x}_k^{\text{new}} - \mathbf{x}_k\|^2 \right\}$$

such that the constraint is satisfied.

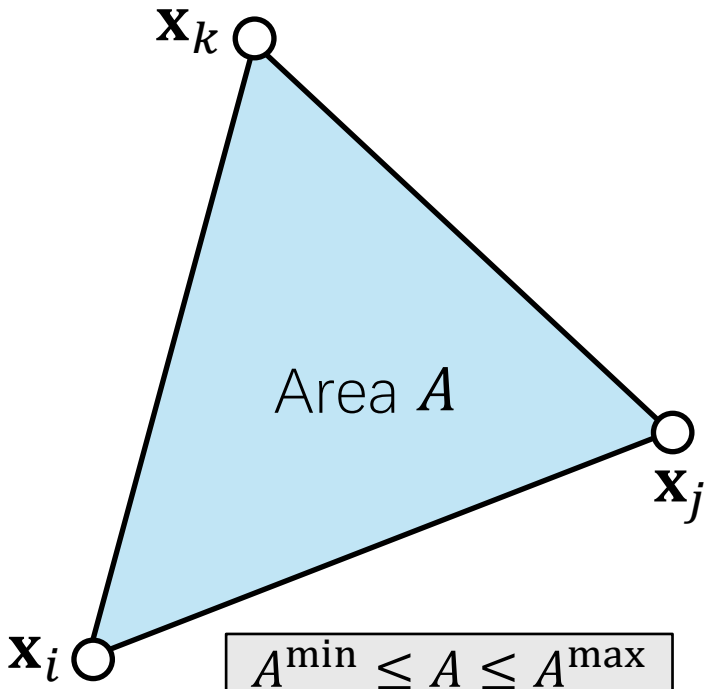


Triangle Area Limit

We can limit the triangle area as well. To do so, we define a scaling factor.

$$\{\mathbf{x}_i^{\text{new}}, \mathbf{x}_i^{\text{new}}, \mathbf{x}_k^{\text{new}}\} = \text{argmin}\frac{1}{2}\left\{m_i\|\mathbf{x}_i^{\text{new}} - \mathbf{x}_i\|^2 + m_j\left\|\mathbf{x}_j^{\text{new}} - \mathbf{x}_j\right\|^2 + m_j\|\mathbf{x}_k^{\text{new}} - \mathbf{x}_k\|^2\right\}$$

such that the constraint is satisfied.



$A^{\min} \le A \le A^{\max}$

Constraint

$$\mathbf{x}^{\text{new}} \longleftarrow \text{Projection}(\mathbf{x})$$

$$A \longleftarrow \frac{1}{2}\left\|(\mathbf{x}_j - \mathbf{x}_i) \times (\mathbf{x}_k - \mathbf{x}_i)\right\|$$

$$s \longleftarrow \sqrt{\min(\max(A, A^{\min}), A^{\max})/A}$$

$$\mathbf{c} \longleftarrow \frac{1}{m_i+m_j+m_k}(m_i\mathbf{x}_i + m_j\mathbf{x}_j + m_k\mathbf{x}_k)$$

$$\mathbf{x}_i^{\text{new}} \longleftarrow \mathbf{c} + s(\mathbf{x}_i - \mathbf{c})$$

$$\mathbf{x}_j^{\text{new}} \longleftarrow \mathbf{c} + s(\mathbf{x}_j - \mathbf{c})$$

$$\mathbf{x}_k^{\text{new}} \longleftarrow \mathbf{c} + s(\mathbf{x}_k - \mathbf{c})$$